Generalization of Church-Rosser Strategies for Confluent Abstract Rewriting Systems of the Smallest Uncountable Cardinality

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Abstract. In the paper we investigate reduction strategies for proving convertibility in confluent abstract rewriting systems (ARS) of the smallest uncountable cardinality that generalize Church-Rosser strategies that are known to exist for countable confluent ARS. Under the Continuum Hypothesis (that is independent of ZFC set theory, if ZFC is consistent), the results obtained in the paper may be useful for guiding the design of (semi-)decision procedures for (restricted instances of) word problems for confluent ARS of the cardinality of the continuum that include rewriting systems on many structures of interest to computer science and applied mathematics (e.g. on the set of graphs with a fixed countable set of vertices, the set of real numbers, the set of finite-dimensional real vectors or matrices, etc.). Proofs of the main results were formalized using Isabelle proof assistant and are available in the supplementary material.

Keywords: Abstract rewriting system · Confluence · Reduction strategy · Word problem · Formal methods · Proof assistant.

1 Introduction

A *word problem* [7,22,2,19] asks whether two mathematical expressions are equivalent in the sense of a specific formal system. Rigorously formulated variants of a word problem have been studied during most of XX century [7,22] (beginning with the works by A. Thue and M. Dehn [22]) for different classes of algebraic structures and rewriting systems. Many of such problem variants turned out to be algorithmically undecidable in the general case, but some turned out to be decidable [7,22]. A highly abstract formulation of a word problem is considered in the *abstract rewriting theory* [2,10,32] inspired by the work [23] by M.H.A. Newman [4]. Abstract rewriting theory can be considered as a part of foundations for more concrete theories of rewriting and applications of rewriting to modeling and analysis of processes of computation and reasoning in different domains, e.g. [32,19,21,30,25].

A word problem for an *abstract rewriting system* (ARS) $(X, \to)$ [2,19] asks for any given elements $a, b \in X$ whether $a \leftrightarrow^* b$ holds (where $\leftrightarrow = \to \cup \to^{-1}$ and $*$ is used to denote the reflexive-transitive closure of a binary relation),

i.e. whether a, b belong to the equivalence relation generated by the reduction relation $\rightarrow$. The latter equivalence is also called a *convertibility* or *conversion* relation [15,5]. At this level of abstraction, the most well known condition that indicates that a word problem may be (in principle) tractable is the condition of *completeness* (confluence and strong normalization) of an ARS.

Completeness ensures that, in principle, the equivalence $a \leftrightarrow^* b$ can be checked [15] by randomly following the reduction relation $\rightarrow$ from the elements a and b till their normal forms (elements that cannot be reduced) and checking the equality of the normal forms of a and b (however, practical feasibility of this procedure depends on representation of elements of an ARS, effectiveness of reduction evaluation and equality checking, and other aspects that are out of scope of the abstract rewriting theory).

The latter fact can be generalized to non-terminating confluent ARS using the notion of a *Church-Rosser strategy* [16,32]. By adapting [16, Definition 6], we will say that a one-step Church-Rosser strategy for an ARS $(X, \rightarrow)$ is a functional subrelation $\rightarrow_s \subseteq \rightarrow$ such that the ARS $(X, \rightarrow)$ and $(X, \rightarrow_s)$ have the same set of normal forms and for every $a, b \in X$, if $a \leftrightarrow^* b$, then $\exists c \in X\ (a \rightarrow_s^* c \wedge b \rightarrow_s^* c)$. This implies that given $\rightarrow_s$ and a, b, if the equivalence $a \leftrightarrow^* b$ holds, then it can be (in principle) proved by (deterministically) following the relation $\rightarrow_s$ from the elements a and b, recording visited elements, and finding a common visited element c (reachable from both a and b) that will be called a *common descendant* [27] of a and b (it also has a role of a witness for $a \leftrightarrow^* b$).

Note that in a (functional) one-step Church-Rosser strategy $\rightarrow_s$ every element has no more than countably many descendants (elements of a reduction sequence $a \rightarrow_s a_1 \rightarrow_s a_2 \rightarrow_s \ldots$), so for a fixed a common descendants of a and b are restricted to a fixed countable set independent of b, which implies that such strategies have limited applicability to uncountable ARS. More precisely, in terms mentioned above, every *countable* confluent ARS *has* a one-step Church-Rosser strategy, but there exist *uncountable* confluent ARS that have *no* one-step Church-Rosser strategy (Proposition 2 in Section 2).

The above mentioned facts motivate investigation of generalizations of one-step Church-Rosser strategies that can provide a method for proving $a \leftrightarrow^* b$ (when it holds) for uncountable confluent ARS (here a proof method is understood in abstract sense at the level of abstract rewriting theory, practical uses of such methods require application-specific representation of elements and reduction rules). In this paper we consider this question for ARS of the smallest uncountable cardinality ($\aleph_1$) in the setting of Zermelo-Fraenkel set theory with the Axiom of Choice (ZFC). More precisely, our results are formalized in the Isabelle proof assistant using HOL logic [24,28], so they can be transferred to ZFC.

The idea of the proposed generalization is to relax the requirement of a Church-Rosser strategy to be deterministic, but restrict nondeterminism of a strategy so that its nondeterministic branchings are used in a proof of $a \leftrightarrow^* b$ as rarely as possible. The topic of this paper is related to the question of what spanning structure can be guaranteed to exist in a connected component of a

confluent ARS of the cardinality $\aleph_1$. We will use technical results on this question from the paper [12] on completeness of the decreasing diagrams method for proving confluence of ARS of the cardinality $\aleph_1$.

Under the Continuum Hypothesis (that is independent of ZFC set theory, if ZFC is consistent), the results obtained in the paper also hold for ARS of the cardinality of the continuum, so they may be useful for designing (semi-)decision procedures for (restricted instances of) word problems for special classes of confluent ARS of the cardinality of the continuum that include rewriting systems on many structures of interest to computer science and applied mathematics (e.g. ARS on the set of graphs with a fixed countable set of vertices, the set of real numbers, the set of finite-dimensional real vectors or matrices, etc.).

2 Preliminaries

A reader can assume that a background theory for the definitions and facts given below is Zermelo-Fraenkel set theory with the Axiom of Choice (ZFC) [8]. Proofs of auxiliary propositions formulated in the paper were formalized and machine-checked in the Isabelle proof assistant [1,24] using HOL logic [28,24]. Results formalized using HOL logic can be translated to the set theoretic setting using a process similar to the one described in [17,18,28].

A short reference table that describes notation used in Isabelle formalizations can be found in [12, Appendix A, table A.1]. A detailed description of Isabelle language can be found in documentation included in Isabelle software [1].

In the paper we will use the following notation:

- $\neg$, $\vee$, $\wedge$, $\Rightarrow$ denote logical connectives (negation, disjunction, conjunction, implication)
- $\mathbb{N}_0 = \{0, 1, 2, ...\}$ is the set of all non-negative integer numbers
- $\mathbb{Z}$ is the set of all integer numbers
- $\mathbb{R}$ is the set of all real numbers
- $|A|$ denotes the cardinality of a set A
- r^+ denotes the transitive closure of a binary relation r
- $r^=$, r^* denote the reflexive closure and the reflexive transitive closure of a binary relation r respectively (on a set that will be clear from the context)
- $r^{-1} = \{(y, x) \mid (x, y) \in r\}$
- $dom(r) = \{x \mid \exists y\ (x, y) \in r\}$ is the domain of a binary relation r.

Sometimes we will use the infix notation to denote that a pair (x, y) belongs to a binary relation R, for example: xRy.

We call an *abstract rewriting system* (ARS) [2,10,32,9] a pair $(X, \rightarrow)$, where X is a set and $\rightarrow \subseteq X \times X$. For convenience and compatibility with HOL formalization [13] (where types cannot be empty), throughout the paper we will implicitly assume that the set of elements of any ARS is *nonempty* ($X \neq \emptyset$), but the main propositions of the paper can be trivially extended to the case of empty ARS.

In an ARS $(X, \to)$ we will call $\to$ a *reduction relation* [2,9,29,10,32] and will call pairs $(a, b) \in \to$ *reduction steps*.

We will say that an ARS $(X, \to)$ is

- *countable* [6], if X is an at most countable set (i.e. there exists a surjective function from the set of natural numbers to X)
- *strongly normalizing* [10], if there is no infinite sequence of reductions

$$x_1 \to x_2 \to x_3 \to ...,$$

  where $x_i \in X$ for $i = 1, 2, 3, ...$
- *confluent* [10], if $\forall a, b, c \in X (a \to^* b \wedge a \to^* c \Rightarrow (\exists d \in X\ (b \to^* d \wedge c \to^* d))$
- *complete* [10], if it is confluent and strongly normalizing.

For an ARS $(X, \to)$ we will usually denote as $\leftrightarrow$ the symmetric closure of the relation $\to$ (i.e. $\leftrightarrow = \to \cup \to^{-1}$), and as $\leftrightarrow^*$ the reflexive transitive closure of $\leftrightarrow$ on X. Note that $\leftrightarrow^*$ is the equivalence relation generated by $\to$ [19], and $\leftrightarrow^*$ is also called the *convertibility* or *conversion* relation [15,5].

An (abstract) *word problem* for an ARS $(X, \to)$ [2,19] asks whether for any given elements $x, y \in X$ the relation $x \leftrightarrow^* y$ holds.

We will use the notation $NF(X, \to)$ for the set of all normal forms [15] (elements that cannot be reduced) in an ARS $(X, \to)$, i.e.

$$NF(X, \to) = X \backslash dom(\to).$$

We will also use the following definitions.

Definition 1 (based on [33, Definition 1]).
A one-step (reduction) strategy for an ARS $(X, \to)$ is a subrelation $\to_s \subseteq \to$ such that $NF(X, \to_s) = NF(X, \to)$, or, equivalently, if $dom(\to) = dom(\to_s)$.

Definition 2. *A one-step strategy $\to_s$ for an ARS $(X, \to)$*

- *is deterministic [16, subsection 2.3], if $\to_s$ is a functional relation, i.e.*

$$\forall a, b\ (a \to_s b \wedge a \to_s c \Rightarrow b = c)$$

- *preserves word problem (or preserves convertibility or preserves conversion), if $\leftrightarrow_s^* = \leftrightarrow^*$ (here the symbol $*$ is used to denote the reflexive transitive closure of a relation on the set X specifically).*

Definition 3 (based on [16, Definition 6]).
A one-step Church-Rosser (or 1-CR) strategy for an ARS $(X, \to)$ is a deterministic one-step strategy $\to_s$ for $(X, \to)$ such that

$$\forall a, b \in X\ (a \leftrightarrow^* b\ \Rightarrow\ \exists c \in X\ a \to_s^* c \wedge b \to_s^* c).$$

Remark 1. We will say that an ARS $(X, \to)$ *has* (or *has no*) strategy with particular properties (e.g. a 1-CR strategy), if there exists (or does not exist) a strategy with the given properties for $(X, \to)$.

Basic facts about 1-CR strategies are summarized in Proposition 1 below (in particular, item 3 of Proposition 1 below highlights a difference between *terminating* confluent (complete) and *non-terminating* confluent ARS).

Proposition 1.

1. *A relation $\to_s$ is a 1-CR strategy for an ARS $(X, \to)$ if and only if $\to_s$ is a deterministic one-step strategy for $(X, \to)$ that preserves word problem.*
2. *In a complete ARS every deterministic one-step strategy is a 1-CR strategy.*
3. *There exists an (acyclic) countable confluent ARS where not every deterministic one-step strategy is a 1-CR strategy.*

Proof (sketch). A machine-checked formal proof in Isabelle is available in [13, lines 2628–2634] (note that in [13] a relation $\to$ is denoted as r, a strategy $\to_s$ is denoted as s, and a nonempty set X implicitly corresponds to the type of components of elements of the relation r).

Proofs of items 1, 2 are straightforward and follow from definitions.

Item 3 can proved using an acyclic, countably infinite ARS illustrated in Figure 1 and formalized in [13, lines 641–646]. Alternatively, item 3 *without* the acyclicity requirement can be proved using a finite ARS with a cycle $(X, \to)$, where $X = \{1, 2\}$ and $a \to b \Leftrightarrow (a = 1 \wedge b \in X)$. □

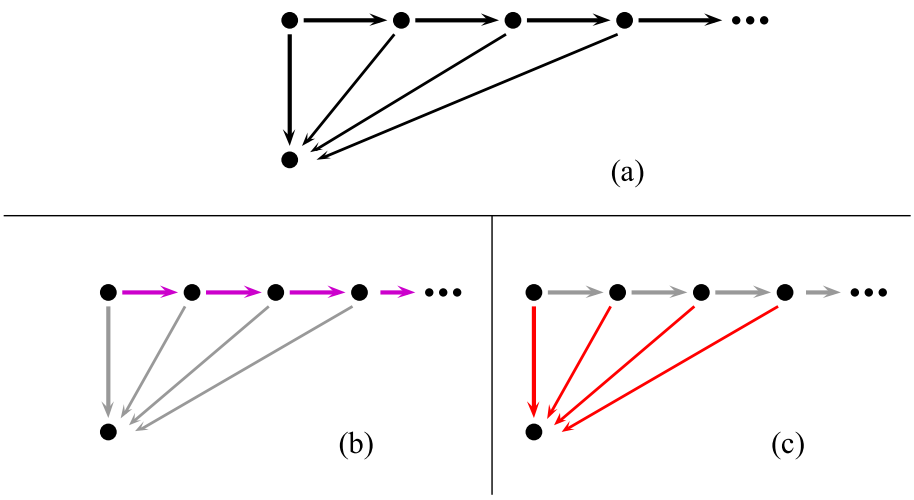

Fig. 1. An illustration for the proof of item 3 of Proposition 1. Subfigure (a): an example acyclic countable confluent ARS (dots denote ARS elements, black arrows denote reduction steps); subfigure (b): violet arrows denote a deterministic one-step strategy that is *not* a 1-CR strategy; subfigure (c): red arrows denote a 1-CR strategy.

Main relations between 1-CR strategies and the confluence property of an ARS are summarized in Proposition 2 below.

Proposition 2.

1. *A countable ARS is confluent if and only if it has a 1-CR strategy.*
2. *Every ARS that has a 1-CR strategy is confluent.*
3. *There exists an (uncountable) confluent ARS* $(X, \to)$ *such that* $|\to| = \aleph_1$ *and* $(X, \to)$ *has no 1-CR strategy.*

Proof (sketch). A machine-checked formal proof in Isabelle is available in [13, lines 2636–2642].

The formal proof of the "only if" part of item 1 is based on construction of a spanning (directed) pseudotree in each connected component of a countable ARS (here a directed pseudotree is a weakly connected directed graph where every node has at most one outgoing arc). The union of such pseudotrees forms a 1-CR strategy as illustrated in Figure 2.

The "if" part of item 1 and item 2 are straightforward and follow from definitions.

Item 3 can be proved by considering a widely known example [6, Footnote 4] of an ARS $(X, \to)$ that lacks the cofinality property [6,14]: X is taken to be the set of all finite subsets of ω_1 (the first uncountable ordinal) and $\to$ is defined by the rule $a \to b \cup \{x\}$ for all $a, b \in X$, $x \in \omega_1$. Then $(X, \to)$ is confluent, but has no 1-CR strategy. □

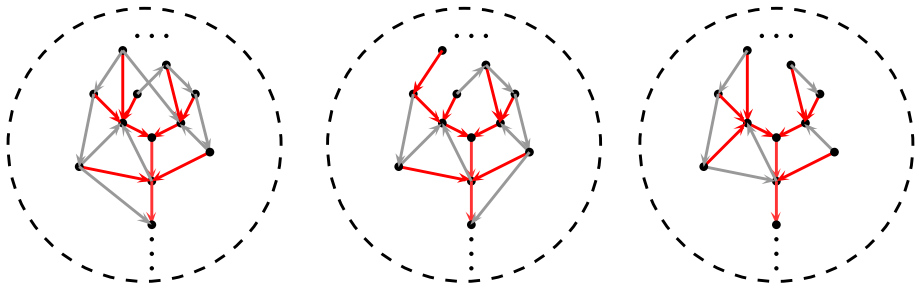

Fig. 2. An illustration for item 1 of Proposition 2. Dashed circles enclose connected components of a countable confluent ARS (considered as a directed graph). Dots denote elements of the ARS. Red and grey arrows denote reduction steps in the ARS. Red arrows denote reduction steps that belong to (possibly infinite) spanning (directed) pseudotrees that together form a 1-CR strategy for the ARS.

3 Generalization of one-step Chuch-Rosser strategies

From item 3 of Proposition 2 it follows that usual (deterministic) one-step Church-Rosser strategies have limited applicability to the problem of proving $a \leftrightarrow^* b$ in uncountable confluent ARS.

We propose an approach that allows one to overcome this issue by relaxing the requirement of a strategy to be deterministic. We will allow elements to have more than one direct successor in a strategy, e.g. $x \to_s y_1$ and $x \to_s y_2$ with $y_1 \neq y_2$, and will assume that traversal of descendants of a and b in such a nondeterministic strategy follows the breadth-first search principle (direct successors of a current element are visited before its further descendants).

However, since nondeterminism of a strategy may complicate search, we will aim to restrict its use. More specifically, in Definition 5 below (of a *k-branching 1-nCR strategy*) we will formalize a restriction that a strategy $\to_s$ is such that reduction steps $x \to_s y$ that have nondeterministic alternatives $x \to_s y' \neq y$ do *not* need to be followed more than k times during traversals of descendants of a and of b in search of a common descendant of a, b.

Let $\to_s$ be a one-step strategy for an ARS $(X, \to)$.

Firstly, let us adapt a few standard concepts from the graph theory to our setting as follows.

Definition 4. *1. A weight function for $\to_s$ is any function $w : (\to_s) \to \mathbb{N}_0$, i.e. w maps arcs $(a, b) \in \to_s$ in the directed graph $(X, \to_s)$ to non-negative integer numbers, so it makes $(X, \to_s)$ a weighted directed graph.*

2. The w-length of a (finite) reduction sequence $a_0 \to_s a_1 \to_s \ldots \to_s a_n$ in $(X, \to_s)$, where w is a weight function for $\to_s$, is the value $\sum_{i=0}^{n-1} w((a_i, a_{i+1}))$.

3. The ball of radius $k \in \mathbb{N}_0$ in $(X, \to_s)$ at an element $a \in X$ w.r.t. a weight function w (for $\to_s$) is the set of all elements $x \in X$ such that there exists a finite reduction sequence $a_0 \to_s a_1 \to_s \ldots \to_s a_n$ (for some $n \in \mathbb{N}_0$) such that its w-length does not exceed k and $a_0 = a$, $a_n = x$.

We will use a particular family of weight functions, parameterized by one-step strategies $\to_s$,

$$bw_{\to_s} : (\to_s) \to \mathbb{N}_0$$

that we will call *branching weight functions*.

For every one-step strategy $\to_s$ for the ARS $(X, \to)$ the weight function $bw_{\to_s}$ is defined as follows:

$$bw_{\to_s}((a, b)) = \begin{cases} 0, & \text{if } \{b' \mid a \to_s b'\} = \{b\}, \\ 1, & \text{if } \{b' \mid a \to_s b'\} \neq \{b\}, \end{cases}$$

i.e. $bw_{\to_s}$ maps a pair $(a, b) \in \to_s$ to the number 0, if b is a unique direct successor of a in $\to_s$, and maps (a, b) to 1 otherwise.

We propose the following generalization of 1-CR strategies.

Definition 5. *For $k \in \mathbb{N}_0$, a k-branching one-step nondeterministic Church-Rosser (or k-branching 1-nCR) strategy for an ARS $(X, \to)$ is a one-step strategy $\to_s$ for $(X, \to)$ such that*

$$\forall a, b \in X \ (a \leftrightarrow^* b \Leftrightarrow B_k(a) \cap B_k(b) \neq \emptyset \ ),$$

where $B_k(x)$ denotes the ball of radius k in $(X, \to_s)$ at an element x w.r.t. the weight function $bw_{\to_s}$ (see Figure 3 below).

Remark 2. The sets $B_k(x)$ in Definition 5 can be explicitly expressed as:

$$B_k(x) = \{\ y \in X \mid \exists n \in \mathbb{N}_0\ \exists a_0, a_1, ..., a_n \in X \\ x = a_0 \to_s a_1 \to_s ... \to_s a_n = y\ \wedge \\ \wedge\ |\{i \in \{0, 1, ..., n-1\} \mid \exists b, b'\ (b \neq b' \wedge a_i \to_s b \wedge a_i \to_s b')\}| \leq k\ \}.$$

Remark 3. Definition 5 means that in order to prove $a \leftrightarrow^* b$ using a k-branching 1-nCR strategy $\to_s$, it is sufficient to explore all reduction sequences in $(X, \to_s)$ that begin at a and b and have $bw_{\to_s}$-length no more than k (i.e. contain no more than k reduction steps that start at an element with non-unique successor in $(X, \to_s)$), and find a common descendant of a, b in them.

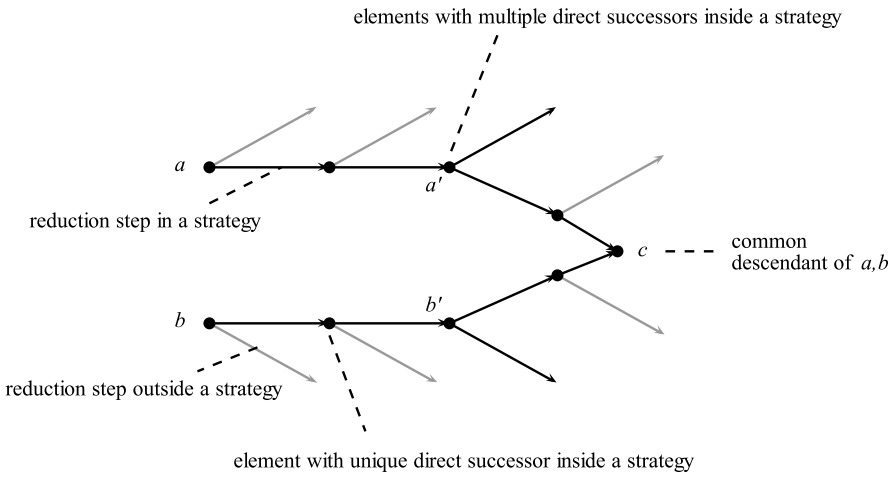

Fig. 3. An illustration for Definition 5. Black dots denote elements of an ARS. Grey and black arrows denote reduction steps in an ARS. Black arrows denote reduction steps that belong to a k-branching 1-nCR strategy in an ARS, grey arrows denote reduction steps that do *not* belong to the given strategy. The k-branching 1-nCR strategy condition: every pair of convertible elements (e.g. a, b in the illustration), can be joined using two reduction sequences that belong to the strategy and contain no more than k reduction steps that originate at elements with non-unique direct successor in the strategy (e.g. a', b' in the illustration).

Proposition 3. *For any ARS $(X, \to)$ and any $k \in \mathbb{N}_0$, every k-branching 1-nCR strategy in $(X, \to)$ preserves word problem.*

A machine-checked formal proof of Proposition 3 in Isabelle is available in [13, lines 2644–2662]. The proof is straightforward and follows from definitions. □

A link between the usual 1-CR strategies and 1-nCR strategies (that are intended to generalize 1-CR strategies) is described in the following proposition.

Proposition 4. *Let $(X, \to)$ be an ARS. Then the following two conditions are equivalent for any relation $\to_s \subseteq X \times X$:*

1. *$\to_s$ is a 1-CR strategy for $(X, \to)$*
2. *$\to_s$ is a deterministic 0-branching 1-nCR strategy for $(X, \to)$.*

A machine-checked formal proof of Proposition 4 in Isabelle is available in [13, lines 2664–2679]. The proof is straightforward and follows from definitions. □

Remark 4. A 0-branching 1-nCR strategy is *not* required to be deterministic (for example, in the ARS $(\{0, 1, 2\}, \to)$, where $\to = \{(0, 1), (1, 2), (2, 1), (1, 0)\}$, the relation $\to_s = \to$ is a nondeterministic 0-branching 1-nCR strategy). However, the Axiom of Choice implies that every ARS has a deterministic one-step strategy, and, in particular, if $\to_s$ is a 0-branching 1-nCR strategy in an ARS $(X, \to)$, then $(X, \to_s)$ has some deterministic one-step strategy $\to'_s \subseteq \to_s$. It is straightforward to check that any such $\to'_s$ is a 1-CR strategy in $(X, \to)$. Thus whenever an ARS $(X, \to)$ has a 0-branching 1-nCR strategy, it also has a 1-CR strategy.

The following analog of item 1 of Proposition 2 holds for 1-nCR strategies.

Proposition 5. *Let $(X, \to)$ be a countable ARS.*
Then $(X, \to)$ is confluent if and only if $(X, \to)$ has a 0-branching 1-nCR strategy.

Proof (sketch). A machine-checked formal proof in Isabelle is available in [13, lines 2681–2685]. The "if" part is straightforward and follows from definitions. The "only if" part follows from Proposition 4 and item 1 of Proposition 2. □

4 Main result

Below we give the main theorem of this paper that is a direct analog of Proposition 5 (applicable to countable ARS) for ARS that have a reduction relation of the cardinality $\aleph_1$ or less (where the "0-branching" condition is replaced with the "1-branching" condition).

Theorem 1. *Let $(X, \to)$ be an ARS such that $|\to| \leq \aleph_1$.*
Then $(X, \to)$ is confluent if and only if $(X, \to)$ has a 1-branching 1-nCR strategy.

Proof (sketch). A machine-checked formal proof in Isabelle is available in [13, line 2693 and below] (note that in [13] a relation $\to$ is denoted as r, a strategy is denoted as s, and a nonempty set X implicitly corresponds to the type of components of elements of r).

The "if" part is straightforward and follows from definitions.

The proof of the "only if" part is based on the idea described below.

Consider a fixed set $U \neq \emptyset$ and a positive integer k. Denote as 2^{U^k} the set of all k-ary relations on U and as $\mathcal{P}_f(A)$ the set of all finite subsets of a set A.

Let us say that a set $\mathcal{R} \subseteq 2^{U^k}$ is *characterized by finite extensions*, if for every $A \in 2^{U^k}$,

$$A \in \mathcal{R} \;\Leftrightarrow\; \mathcal{P}_f(A) \cap \mathcal{R} \text{ is a cofinal subset of } \mathcal{P}_f(A) \text{ w.r.t. } \subseteq$$

i.e.

$$A \in \mathcal{R} \;\Leftrightarrow\; (\forall F \in \mathcal{P}_f(A) \;\; \exists F' \in \mathcal{P}_f(A) \cap \mathcal{R} \quad F \subseteq F').$$

Confluence is an example of a property of interest to the abstract rewriting theory that induces a set characterized by finite extensions: using induction on the cardinality of a finite subrelation of a confluent relation it can be shown that the set $CR(U) = \{r \in 2^{U \times U} \mid (U, r) \text{ is a confluent ARS}\}$ is characterized by finite extensions.

Let us say that a set $\mathcal{R} \subseteq 2^{U^k}$ is *closed under chain unions*, if for every $\mathcal{C} \subseteq \mathcal{R}$:

$$(\forall A, B \in \mathcal{C} \;\; A \subseteq B \vee B \subseteq A) \;\;\Rightarrow\;\; \left(\bigcup \mathcal{C}\right) \in \mathcal{R}.$$

Also, let us say that a set $\mathcal{R}' \subseteq 2^{U^k}$ is the *chain-union closure* of $\mathcal{R} \subseteq 2^{U^k}$, if $\mathcal{R}'$ is the least (in the sense of $\subseteq$) subset of 2^{U^k} such that

$$\mathcal{R} \subseteq \mathcal{R}' \text{ and } \mathcal{R}' \text{ is closed under chain unions.}$$

Using an argument reminiscent to the proof of Iwamura's lemma about directed sets [20] it can be shown that if a set $\mathcal{R} \subseteq 2^{U^k}$ is characterized by finite extensions and contains the empty relation (i.e. $\emptyset \in \mathcal{R}$), then $\mathcal{R}$ is the chain-union closure of $\{A \in \mathcal{R} \mid A \text{ is finite}\}$.

In particular, since $CR(U)$ is characterized by finite extensions and $\emptyset \in CR(U)$, the set $CR(U)$ is the chain-union closure of $\{r \in CR(U) \mid r \text{ is finite}\}$. This statement can be viewed as an induction principle for proving completeness of confluence criteria. However, we do not apply it directly to the "only if" part of the current theorem. Instead, we constructed a refinement of such an induction principle (in essence, by imposing a number of technical conditions on elements of a chain that approximate a confluent relation) that implies that in a confluent relation $\rightarrow$ of cardinality $\aleph_1$ a connected component that lacks the cofinality property can be represented as a union of a chain of countable connected components of confluent relations that has a spanning subgraph of the form illustrated in Figure 4 (formed by red and green arrows). The cardinality assumption $|\rightarrow| \leq \aleph_1$ is used in the proof to guarantee countability of members of this chain. Then the union of such spanning subgraphs for connected components of an ARS that lack the cofinality property and of spanning (directed) pseudotrees for components that have the cofinality property is shown to form a 1-branching 1-nCR strategy.

Details of the proof are available in formal lemmas *lem_ex1b1CR_confl_w1* and *lem_exspn_1ndccr_refl_ccr_scfw1* at lines 2485–2550 and 2051–2389 in [13].

□

Corollary 1. *In ZFC + CH (Continuum Hypothesis), every confluent ARS* $(X, \rightarrow)$ *with* $|\rightarrow| \leq |\mathbb{R}|$ *has a 1-branching 1-nCR strategy.*

Remark 5. In practical sense the CH assumption is not very restrictive, since, assuming ZFC is consistent, for any concrete $(X, \rightarrow)$ it is *not* provable in ZFC that $(X, \rightarrow)$ is a confluent ARS with $|\rightarrow| \leq |\mathbb{R}|$ that lacks a 1-branching 1-nCR strategy.

Remark 6. Elements with non-unique successor in a 1-branching 1-nCR strategy (e.g. a', b' in Figure 3) can be thought of as kinds of "*virtual normal forms*": in Figure 3, a', b' are determined uniquely by a and b respectively (within a strategy). Of course, they are not true normal forms (they are reducible), but Theorem 1 suggests that (when they exist) they can serve as important milestones during search of a common descendant of a, b.

In particular, Theorem 1 suggests that when a relation $\rightarrow$ of cardinality $\aleph_1$ (or cardinality $|\mathbb{R}|$ under CH) in an abstract word problem $a \leftrightarrow^* b$ is known to be confluent (or can be enforced to be confluent like in completion procedures, e.g. using the decreasing diagrams method [26] for analysis of local peaks), when designing a (semi-)decision procedure for (possibly restricted instances of) this problem, one can try to:

1. select a set of elements that will serve as virtual normal forms (elements with non-unique successor in a 1-branching 1-nCR strategy)
2. design a deterministic procedure to reach them from a given element (perform "*virtual normalization*")
3. design a procedure to check joinability of such elements.

Note that for 1-nCR strategies it is *not* required that any given element (e.g. a virtual normal form mentioned above) has finitely many direct successors in a strategy since this would imply a restriction on the cardinality of the set of descendants of any element in a strategy like in the case of deterministic strategies. In fact, in the case of uncountable ARS an element may have uncountably many direct successors in a 1-nCR strategy, so, in practical sense, search in such a strategy should be understood *not* as enumeration of successors of an element, but as logical analysis of properties of successors of an element (e.g. using automated theorem proving techniques).

5 Examples

The examples given below illustrate the notion of a k-branching 1-nCR strategy.

Example 1. Let $(X, \rightarrow)$ be an ARS where $X = \mathbb{R}^2$ (i.e. elements are 2-dimensional real vectors) and $\rightarrow$ is defined by the rule

$$(x, y) \rightarrow (ky, kx) \text{ for every } k \in \mathbb{R} \text{ such that } k > 0,$$

or, more formally,

$$\rightarrow = \{((x, y), (x', y')) \in \mathbb{R}^2 \times \mathbb{R}^2 \mid \exists k \in \mathbb{R}\ k > 0 \wedge x' = ky \wedge y' = kx\},$$

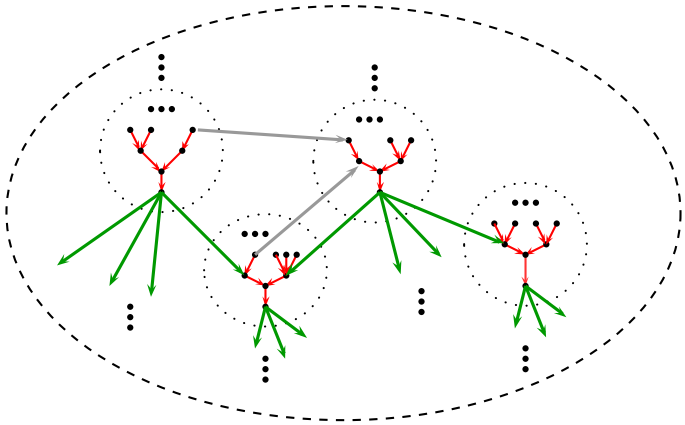

Fig. 4. An illustration for the "only if" part of Theorem 1. Dashed ellipse encloses a connected ARS of the cardinality $\aleph_1$ that lacks cofinality property, where grey, red, green arrows denote reduction steps colored in such a way that, *ignoring* grey arrows, red arrows start at elements with unique direct successor and green arrows start at elements with non-unique direct successor. This connected component lacks a spanning (directed) pseudotree, but has a spanning structure that consists of all red and green arrows that forms a 1-branching 1-nCR strategy for this component (node pairs can be joined without using green arcs more than once on paths to their common descendant).

i.e. components of a vector (x, y) are scaled using a positive real coefficient k and exchanged. It is straightforward to check that $(X, \rightarrow)$ is confluent, but *not* terminating (e.g. $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3) \rightarrow ...$).

Consider a relation $\rightarrow_s \subseteq X \times X$ such that $(x, y) \rightarrow_s (x', y')$ if and only if the following two conditions hold:

1. $(x, y) \rightarrow (x', y')$
2. if $x > y$, then $(x', y') = (y, x)$.

Then $\rightarrow_s$ is 1-branching 1-nCR strategy. Here virtual normal forms mentioned in Remark 6 are vectors (x, y) with $x \leq y$. If a pair of such vectors has a common descendant in $\rightarrow$, it can found immediately in 1 reduction step by finding an appropriate positive scaling coefficient k in the definition of $\rightarrow$ above.

Despite being uncountable, this ARS also has a 0-branching 1-nCR strategy, and, in fact, it has a 1-CR strategy, e.g. $\rightarrow'_s \subseteq X \times X$ such that $(x, y) \rightarrow'_s (x', y')$ if and only if the following two conditions hold, where $||.||$ is the Euclidean norm:

1. if $x > y$ or $||(x, y)|| = 0$, then $(x', y') = (y, x)$
2. if $x \leq y$ and $||(x, y)|| \neq 0$, then $(x', y') = (y, x)/||(y, x)||$.

Still, it follows from item 3 of Proposition 2 and Remark 4 that there *do* exist uncountable ARS that have *no* 0-branching 1-nCR strategy.

Example 2. Assume the Continuum Hypothesis (CH) and consider ARS $(X, \to)$ based on [6, Footnote 4], where X is the set of all finite subsets of $\mathbb{R}$ (i.e. elements of the ARS are finite sets of real numbers) and

$$\to = \{(a, b) \in X \times X \mid \exists x \in \mathbb{R}\ b = a \cup \{x\}\}.$$

Then $(X, \to)$ satisfies the conditions of Theorem 1, is confluent, and has a 1-branching 1-nCR strategy.

The word problem for this ARS is trivial ($a \leftrightarrow^* b$ holds for all $a, b \in X$, i.e. $(X, \to)$ is weakly connected as a graph), but the existence of a 1-branching 1-nCR strategy may not be immediately obvious.

In this example an explicit 1-branching 1-nCR strategy can be given as follows (without assuming CH). Let $F : [0, 1) \to X$ be a surjective function (such a function exists, because $|X| = |\mathbb{R}|$, but for a more explicit construction of F see Remark 7 below).

For every $x \in \mathbb{R}$ denote $F_p(x) = F(x - \lfloor x \rfloor)$, where $\lfloor x \rfloor = \max\{n \in \mathbb{Z} \mid n \leq x\}$.

Let $\to_s \subseteq X \times X$ be a relation such that for all $a, b \in X$, $a \to_s b$ if and only if there exists $x \in \mathbb{R}$ such that the following two conditions hold:

1. $b = a \cup \{x\}$
2. if $a \neq \emptyset$, $F_p(\max a) \backslash a \neq \emptyset$, and $\min(F_p(\max a) \backslash a) < \max a$, then $x = \min(F_p(\max a) \backslash a)$.

Then $\to_s$ is a one-step strategy for $(X, \to)$.

For $k = 0, 1$ denote as $B_k(x)$ the ball of radius k in $(X, \to_s)$ at an element x w.r.t. $bw_{\to_s}$. Let $a, b \in X$. It is straightforward to check that there exist $a' \in B_0(a)$ and $b' \in B_0(b)$ such that $\{c \mid a' \to_s c\} = \{c \mid a' \to c\}$ and $\{c \mid b' \to_s c\} = \{c \mid b' \to c\}$. Since F is surjective, there exists $t \in [0, 1)$ such that $a' \cup b' = F(t)$. Let $u = t + \min\{n \in \mathbb{N}_0 \mid \forall x \in a' \cup b'\ x < n\}$. Then $a' \to_s a' \cup \{u\}$, $b' \to_s b' \cup \{u\}$. Moreover, $u \notin a' \cup b'$ and $u = \max(a' \cup \{u\}) = \max(b' \cup \{u\})$. Then

$$F_p(u) \backslash (a' \cup \{u\}) = F(t) \backslash (a' \cup \{u\}) = (a' \cup b') \backslash (a' \cup \{u\}) = b' \backslash a',$$

$$F_p(u) \backslash (b' \cup \{u\}) = F(t) \backslash (b' \cup \{u\}) = (a' \cup b') \backslash (b' \cup \{u\}) = a' \backslash b'.$$

Then it is straightforward to check that $a' \cup b' \cup \{u\} \in B_0(a' \cup \{u\}) \cap B_0(b' \cup \{u\})$. Then $B_1(a) \cap B_1(b) \neq \emptyset$. From this it follows that $\to_s$ is a 1-branching 1-nCR strategy.

Remark 7. An example of a surjective function F from $[0, 1)$ to the set of all finite subsets of $\mathbb{R}$ mentioned in Example 2 can be described as follows.

Let us fix a surjective function $g : \mathbb{R} \to \mathbb{R}^2$ (g can be required to be *continuous* and can be obtained using one of *space-filling curve* constructions [31]).

For every $x \in \mathbb{R}$ denote as $g_0(x)$ and $g_1(x)$ components of $g(x)$, i.e. real numbers such that $g(x) = (g_0(x), g_1(x))$. For every $x \in \mathbb{R}$ denote

$$F_0(x) = \{g_1(g_0^{(i)}(x)) \mid i \in \mathbb{N} \wedge i \leq g_1(x)\},$$

where $\mathbb{N} = \{1, 2, 3, ...\}$ and $g_0^{(i)}(x) = \underbrace{g_0(g_0(...(g_0(x)...))}_{i}$.

As in Example 2, let X be the set of all finite subsets of $\mathbb{R}$. Then it is straightforward to check that F_0 is a surjective function from $\mathbb{R}$ to X.

Let $f : [0, 1) \to \mathbb{R}$ be a surjective function, e.g.

$$f(t) = \frac{1}{1-t} \sin\left(\frac{1}{1-t}\right)$$

for $t \in [0, 1)$ and let $F : [0, 1) \to X$ be the composition of F_0 and f, i.e.

$$F(t) = F_0(f(t))$$

for all $t \in [0, 1)$. Then F is a surjective function from $[0, 1)$ to X.

6 Formalization in proof assistant

We formalized propositions from Sections 2, 3 and Theorem 1 from Section 4 (main theorem) using Isabelle 2025 proof assistant [1,24] and HOL logic in the file `NCRS_N1.thy` [13] that consists of about 2700 lines. The theory `NCRS_N1.thy` depends on a previous theory `DCRN1.thy` [11] (supplementary material for the paper [12]) and reuses parts of proofs of technical results from `DCRN1.thy`.

Instructions on re-checking formal proofs are given in Appendix at the end of this paper.

Unlike most usual formalizations of results from the rewriting theory in Isabelle/HOL, our formalization significantly uses basic facts about ordinal numbers and cardinalities of sets. For this reason, we use a formalization of ordinal and cardinal numbers *HOL-Cardinals* described in [3] as a dependency.

Formalizations of main definitions from Sections 2, 3 in [13, lines 25–65] have the following forms:

— *s* is a one-step strategy for ARS *(U,r)*

```
abbreviation is_1step_strategy :: "'U rel ⇒ 'U rel ⇒ bool"
where
  "is_1step_strategy s r ≡ s ⊆ r ∧ Domain s = Domain r"
```

— *s* is a deterministic one-step strategy for ARS *(U,r)*

```
definition is_det_1step_strategy :: "'U rel ⇒ 'U rel ⇒ bool"
where
  "is_det_1step_strategy s r ≡ is_1step_strategy s r ∧ single_valued s"
```

— *s* is a one-step Church-Rosser strategy for ARS *(U,r)*

```
definition is_CR_1step_strategy :: "'U rel ⇒ 'U rel ⇒ bool"
where
  "is_CR_1step_strategy s r ≡ is_det_1step_strategy s r
                          ∧ ( ∀ a b. (a,b) ∈ (r ∪ r^-1)^* ⟶
                                (∃ c. (a,c) ∈ s^* ∧ (b,c) ∈ s^*) )"
```

— *ai* is a (finite) reduction sequence of *n* elements in ARS *(U,s)*

```
definition is_finrseq :: "'a rel ⇒ nat ⇒ (nat ⇒ 'a) ⇒ bool"
where
  "is_finrseq s n ai ≡ (∀ i < n. (ai i, (ai (Suc i))) ∈ s)"
```

— *w*-length of a reduction sequence *ai* of *n* elements

```
definition wlen :: "('a ⇒ 'a ⇒ nat) ⇒ (nat ⇒ 'a) ⇒ nat ⇒ nat"
where
  "wlen w ai n ≡ ( ∑i<n. w (ai i) (ai (Suc i)) )"
```

— ball of radius *k* at *a* in ARS *(U,s)* w.r.t. weight function *w*

```
definition wball :: "'a rel ⇒ ('a ⇒ 'a ⇒ nat) ⇒ nat ⇒ 'a ⇒ 'a set"
where
  "wball s w k a = {x. ∃ n ai. is_finrseq s n ai
                          ∧ wlen w ai n ≤ k ∧ ai 0 = a ∧ ai n = x }"
```

— *(bw s)* is a branching weight function for ARS *(U,s)*

```
definition bw :: "'a rel ⇒ 'a ⇒ 'a ⇒ nat"
where
  "bw s a b ≡ (if {b'. (a,b') ∈ s} = {b} then 0 else 1)"
```

— *s* is a *k*-branching one-step nondeterministic Church-Rosser strategy in ARS *(U,r)*

```
definition is_kb_1nCR_strategy :: "nat ⇒ 'U rel ⇒ 'U rel ⇒ bool"
where
  "is_kb_1nCR_strategy k s r ≡ is_1step_strategy s r
      ∧ ( ∀ a b. (a,b) ∈ (r ∪ r^-1)^* ⟷
                 wball s (bw s) k a ∩ wball s (bw s) k b ≠ {} )"
```

Also, in [13] the following formal definition of the confluence property from [11, lines 99–100] is used (here r denotes a binary relation):

```
definition confl_rel
  where "confl_rel r ≡ (∀a b c. (a,b) ∈ r^* ∧ (a,c) ∈ r^* ⟶
                          (∃ d. (b,d) ∈ r^* ∧ (c,d) ∈ r^*) )"
```

Formalization of Proposition 1 in [13, lines 2628–2633] has the following form:

```
lemma lem_prop1:
    "∀ s r. is_CR_1step_strategy s r = (is_det_1step_strategy s r ∧
                                        (s ∪ s^-1)^* = (r ∪ r^-1)^*)"
and "∀ r. ( confl_rel r ∧ wf (r^-1) ⟶
        (∀ s. is_det_1step_strategy s r ⟶ is_CR_1step_strategy s r) )"
and "∃ r::nat rel. ( confl_rel r ∧
    ¬ ( ∀ s. is_det_1step_strategy s r ⟶ is_CR_1step_strategy s r ) )"
```

Formalization of Proposition 2 in [13, lines 2636–2641] has the following form:

```
lemma lem_prop2:
    "∀ r::'a rel. ( |{x::'a. True}| ≤o |{x::nat. True}| ⟶
                    ( confl_rel r = (∃ s. is_CR_1step_strategy s r) ) )"
and "∀ r. ( (∃ s. is_CR_1step_strategy s r) ⟶ confl_rel r )"
```

```
and "∃ r::nat set set rel. |r| =o cardSuc |{x::nat. True}| ∧
                    confl_rel r ∧ ( ¬ (∃ s. is_CR_1step_strategy s r) )"
```

Formalizations of Propositions 3, 4, 5 in [13, lines 2644–2685] have the following forms:

```
lemma lem_prop3:
fixes r::"'U rel" and k::nat
shows "∀ s. is_kb_1nCR_strategy k s r ⟶ (s ∪ s^-1)^* = (r ∪ r^-1)^*"

lemma lem_prop4:
fixes r s::"'U rel"
shows "is_CR_1step_strategy s r =
                  (is_det_1step_strategy s r ∧ is_kb_1nCR_strategy 0 s r)"

lemma lem_prop5:
fixes r::"'U rel"
assumes "|{x::'U. True}| ≤o |{x::nat. True}|"
shows "confl_rel r = (∃ s. is_kb_1nCR_strategy 0 s r)"
```

Formalization of Theorem 1 (main theorem) in [13, lines 2693–2696] has the following form:

```
theorem thm_1:
fixes r::"'U rel"
assumes "|r| ≤o cardSuc |{x::nat. True}|"
shows "confl_rel r = (∃ s. is_kb_1nCR_strategy 1 s r)"
```

Here the symbols $|\cdot|$ and $\leq_0$, and *cardSuc* denote the cardinality of a set/relation, cardinality comparison, and the cardinal successor operation respectively. They are defined in the theory *HOL-Cardinals* [3].

7 Conclusions

In the paper we have investigated strategies for proving convertibility in confluent ARS of the smallest uncountable cardinality that generalize one-step Church-Rosser strategies for countable confluent ARS. Under the Continuum Hypothesis, the results obtained in the paper (in particular, Theorem 1) may be useful for guiding the design of (semi-)decision procedures for (restricted instances of) word problems for confluent ARS of the cardinality of the continuum that include rewriting systems on many structures of interest to computer science and applied mathematics (e.g. on the set of graphs with a fixed countable set of vertices, the set of real numbers, the set of finite-dimensional real vectors or matrices, etc.).

Investigation of applications of the obtained results to concrete rewriting systems and investigation of strategies for ARS of high uncountable cardinality are subjects of future work.

Appendix

Instructions on re-checking formal proofs in the supplementary material [13] are given below:

1. Install Isabelle 2025 software using instructions at:

   https://isabelle.in.tum.de/website-Isabelle2025/installation.html

2. Run the installed Isabelle 2025 software.
3. Obtain the file `NCRS_N1.thy` (supplementary material for this paper) from:

   https://doi.org/10.5281/zenodo.18079151

   and obtain the file `DCRN1.thy` (dependency of `NCRS_N1.thy`) from:

   https://doi.org/10.5281/zenodo.14254256

4. Place both files `NCRS_N1.thy` and `DCRN1.thy` in the same directory.
5. Choose File → Open from the menu in Isabelle window and select the file `NCRS_N1.thy` (its dependency will be loaded automatically).
6. Scroll to the end of the loaded text in Isabelle window and wait until all red marks on the right scroll bar disappear (this may take several minutes).
7. Now formal proofs have been checked. The main result (theorem `thm_1`) is located at the line 2693 in `NCRS_N1.thy`.

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